A radius-one shift with entropy greater than log two

Disproof of Conjecture 00000009624

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Abstract

The bilateral full shift on three symbols is a surjective sliding-block automorphism with minimal radius one. Its actual dynamical cylinder covering number at time n is 3^n, giving topological entropy at least $\log 3 > \log 2$. This violates the proposed bound $h \le r \log 2$ under the source's unrestricted finite-alphabet statement. The Lean proof directly uses Mathlib's topological entropy, not a word-count quantity merely named entropy.

The actual compact system and local rules

Let $\mathcal{A} = \{0, 1, 2\}$ have its discrete topology, and use the full configuration space

$$X = \mathcal{A}^{\mathbb{Z}}$$

with its product topology and product uniformity. It is compact. This is the full shift, also a shift of finite type with an empty list of forbidden words. The exact bilingual source does not restrict the alphabet to two symbols.

Define $F : X \to X$ and $G : X \to X$ by

$$F(x)_k = x_{k+1}, \qquad G(x)_k = x_{k-1}.$$

They are continuous because each output coordinate is a coordinate projection. Direct substitution gives $FG = GF = \mathrm{id}$, so F is a homeomorphism and is surjective. It commutes with the underlying shift, being that shift itself; therefore it is an automorphism of this full shift.

A radius-r sliding-block rule uses the symmetric window $k - r, \ldots, k + r$ to produce output at k. For F, the radius-one rule sends (a, b, c) to c; the inverse rule sends it to a. Thus both directions have radius one.

This is also the true minimum radius. Let x be the all-zero configuration and let y be zero except that $y_1 = 1$. They agree at coordinate zero, but $F(x)_0 = 0$ and $F(y)_0 = 1$. A radius-zero rule could not distinguish them. Hence the minimal radius of F is exactly one, not merely a padded choice of a larger window.

Actual dynamical covering numbers

Use the genuine uniform entourage

$$U = \{(x, y) : x_0 = y_0\}.$$

It is an entourage because the alphabet is discrete and the zero-coordinate projection is uniformly continuous. The actual iterates satisfy $F^j(x)_k = x_{k+j}$ for every $j \geq 0$. Consequently the n-step dynamical ball is exactly

$$B_n(x, U) = \{y : (F^j x, F^j y) \in U \ (0 \leq j < n)\} = \{y : y_j = x_j \ (0 \leq j < n)\}.$$

Thus these dynamical balls are precisely the actual length-n cylinders in coordinates $0, \ldots, n-1$.

Every word $w \in \mathcal{A}^n$ extends to a genuine bilateral configuration by setting all coordinates outside this block to zero. Reading the block back recovers w. Hence all 3^n words occur in the full system, not just in a proposed counting model.

Let $r_n(U)$ be the minimal size of a finite family of such dynamical balls covering all of X. In any cover, each center has one length-n word, and that ball covers only configurations with that same word. Since every word occurs, the map from centers to their words must be surjective; therefore there are at least 3^n centers.

Conversely, choose the zero-extended configuration for every word. These form a cover of all X, and the extension map is injective because reading recovers the word. This cover has exactly 3^n centers. We have proved the genuine minimum

$$r_n(U) = 3^n.$$

The equality even holds at $n = 0$, when one ball covers the nonempty configuration space.

The genuine entropy obstruction

For the compact uniform system, Bowen–Dinaburg topological entropy is

$$h_{\mathrm{top}}(F) = \sup_{V \in \mathcal{U}_X} \limsup_{n \to \infty} \frac{1}{n} \log r_n(V).$$

Applying this definition to the actual entourage U and its exact minimum covering number gives

$$h_{\mathrm{top}}(F) \geq \log 3 > \log 2 = 1 \cdot \log 2.$$

Only this lower bound is needed; the formal proof does not claim an unproved upper bound on the total entropy. All logarithms use the same base, chosen to be the natural base in Lean.

Conclusion, formalization and reproducibility

Theorem 1. *A surjective sliding-block automorphism of minimal radius one need not satisfy* $h \leq \log 2$.

Proof. The actual three-symbol shift F has minimal radius one and genuine topological entropy at least $\log 3 > \log 2$, by the covering argument above. □

This refutes the first asserted bound as written. A binary-alphabet restriction would be an additional hypothesis, absent from both source versions. The equality-family and entropy-spectrum clauses need not be separately refuted once the proposed universal inequality is false.

The Lean project uses the entire function space $\mathbb{Z} \to \mathrm{Fin}(3)$ with its actual discrete-product uniformity. It constructs the homeomorphism and inverse, proves surjectivity, uniform continuity, compactness, the two local rules, shift commutation and the radius-zero obstruction. No finite periodic state space replaces the full system.

It proves the exact iterate formula and identifies actual dynamical balls with true block agreement. Reading and extending words give the lower and upper bounds for Mathlib's genuine dynamical-cover minimum. The library's original exponential-growth and topological-entropy definitions then give the entourage entropy $\log 3$ and total entropy lower bound. A final theorem directly negates the universal radius–entropy inequality for the actual shift-commuting homeomorphisms. No external symbolic entropy bridge is assumed.

The supplementary `verify.py` checks all words of lengths zero through eight, their actual extended orbit traces, both inverse identities and radius-one rules, and the radius-zero witness using exact symbols. These are finite consistency checks; Lean proves the full infinite-space and all-n statements.

Lean 4.19.0 and Mathlib revision
`c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From `lean/`, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

Fresh-build logs, source and hashes are included. Printed dependencies contain only standard foundational axioms. Compile with `tectonic main.tex`. The author agent and parent reviewed the work internally; no external independent review is claimed.

References

[1] The Last Math Competition, Conjecture 00000009624. Exact bilingual source included.

[2] Mathlib, *Topological entropy via covers*, pinned revision above. See the module `CoverEntropy` in the topological entropy library. The project directly uses these definitions and proves its own dynamical covering cardinalities.